Case study 6:-

Weather Monitoring System:- A weather station stores temperature in Celsius but international reports require Fahrenheit. Develop a converter.

Algorithm – Weather Monitoring System

1. Start.
2. Declare variables: celsius, fahrenheit.
3. Input the temperature in Celsius.
4. Calculate Fahrenheit using the formula:
   - fahrenheit = (celsius × 9 / 5) + 32
5. Display the temperature in Fahrenheit.
6. Stop.

Pseudocode:-

BEGIN

DECLARE celsius, fahrenheit AS FLOAT

INPUT celsius

fahrenheit = (celsius * 9 / 5) + 32

PRINT "Fahrenheit = ", fahrenheit

END

Test cases:- input/output combination.

```
> gcc CASE_study_6.c -o CASE_study_6 ; if ($?) { .\CASE_study_6 }
enter the  temperature in celsius 30
 fahrenheit=86.00
PS C:\Users\khush\Desktop\PROJECT 1>
```

```
PS C:\Users\khush\Desktop\PROJECT 1> gcc CASE_study_6.c -o CASE_study_6 ; if ($?) { .\CASE_study_6 }
enter the  temperature in celsius 45
 fahrenheit=113.00
PS C:\Users\khush\Desktop\PROJECT 1>
```

```
PS C:\Users\khush\Desktop\PROJECT 1> gcc CASE_study_6.c -o CASE_study_6 ; if ($?) { .\CASE_study_6 }
enter the  temperature in celsius 50
 fahrenheit=122.00
PS C:\Users\khush\Desktop\PROJECT 1>
```